GEOMETRY QUEST · WORLD 14

Congruence

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

1 Two triangles share these facts. Which rule proves they are congruent: SSS, SAS, ASA, AAS, RHS or “not enough”?

all three sides match	**SSS**
two sides and the angle between them match	**SAS**
two angles and the side between them match	**ASA**
a right angle, the hypotenuse and one other side match	**RHS**
all three angles match	**not enough**
two sides and an angle NOT between them match	**not enough**

2 △ABC ≅ △PQR. AB = 5 cm, BC = 7 cm and angle A = 40°. Find PQ, QR and angle P.

PQ = **5 cm** **QR =** **7 cm** **∠P =** **40°**

3 Draw two triangles that both have angles of 50°, 60° and 70° but are different sizes. What does this show?

Matching angles (AAA) is not enough: same shape, different size (similar).

4 Why does a garden gate have a diagonal piece of wood across it?

The brace makes triangles, which are rigid (SSS), so the gate can’t sag.

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